

Full Stack Development Laboratory 03

Aim: Client-side Form Validations using JavaScript, DOM real-time update, JQuery to develop Ajax based applications.

Aim: Write a program to perform following form validations using JavaScript:

- a) All fields mandatory,
 - b) Phone number, Email Address, Zip code Validation etc.
- Include JavaScript to access and manipulate Document Object Model (DOM) objects in an HTML web page.
Include JQuery to develop to develop your application as an Ajax based application.

Objectives:

1. To understand what form validation is.
2. To learn basic functioning of DOM objects.
3. To learn how to apply various techniques to implement it.

Theory:

Different types of form validations.

Form validation is the process of checking whether the data entered by a user in a form is correct, complete, and acceptable before submission. It helps maintain data accuracy and security.

Types of Form Validations

1. **Client-Side Validation**
This validation is performed in the user's browser using HTML, CSS, or JavaScript before sending data to the server. It provides quick feedback and reduces server load.
2. **Server-Side Validation**
This validation is done on the server after the form is submitted. It ensures data security and correctness even if client-side validation is bypassed.
3. **Required Field Validation**
Ensures that mandatory fields are not left empty, such as name, email, or password.
4. **Format Validation**
Checks whether the data is in the correct format, for example, valid email address, phone number, or date format.
5. **Length Validation**
Verifies that input data meets minimum or maximum length requirements, such as password length.

HTML Document Object Model.

The HTML Document Object Model (DOM) is a programming interface that represents an HTML document as a structured tree of objects. Each element, attribute, and piece of text in the document is treated as a node in this tree.

The DOM allows programs, especially JavaScript, to access, modify, add, or delete HTML elements dynamically. This makes web pages interactive by enabling changes in content, style, and structure without reloading the page.

In simple terms, the DOM acts as a bridge between HTML documents and scripting languages, allowing developers to manipulate web pages easily.

What is JQuery? Write various JQuery Selectors.

jQuery is a fast, lightweight JavaScript library designed to simplify HTML document manipulation, event handling, animations, and AJAX interactions. It makes it easier to write JavaScript code by providing simple methods to select and manipulate HTML elements.

Various jQuery Selectors

jQuery selectors are used to select and manipulate HTML elements on a web page.

1. **Element Selector** – Selects elements by tag name.
Example: \$("p") → Selects all paragraph elements.
2. **ID Selector** – Selects an element using its unique ID.
Example: \$("#header") → Selects element with ID *header*.
3. **Class Selector** – Selects elements with a specific class.
Example: \$(".menu") → Selects all elements with class *menu*.
4. **Universal Selector** – Selects all elements.
Example: \$("*").
5. **Attribute Selector** – Selects elements based on attributes.
Example: \$("[type='text']").

FAQ:

1. Write 3 reasons why Form validations are important.

Three Reasons Why Form Validations Are Important

1. **Ensures Accurate Data**
Form validation helps ensure that users enter correct and complete information, reducing errors in the submitted data.
2. **Improves Security**
It prevents invalid or harmful data from being submitted, protecting websites from security threats such as malicious inputs.
3. **Enhances User Experience**
Validation provides immediate feedback to users, helping them correct mistakes quickly and making forms easier to use.

2. Give an example of how to modify an attribute value using DOM.

The HTML Document Object Model (DOM) allows JavaScript to modify attributes of HTML elements dynamically. This can be done using the `setAttribute()` method.

Example:

```

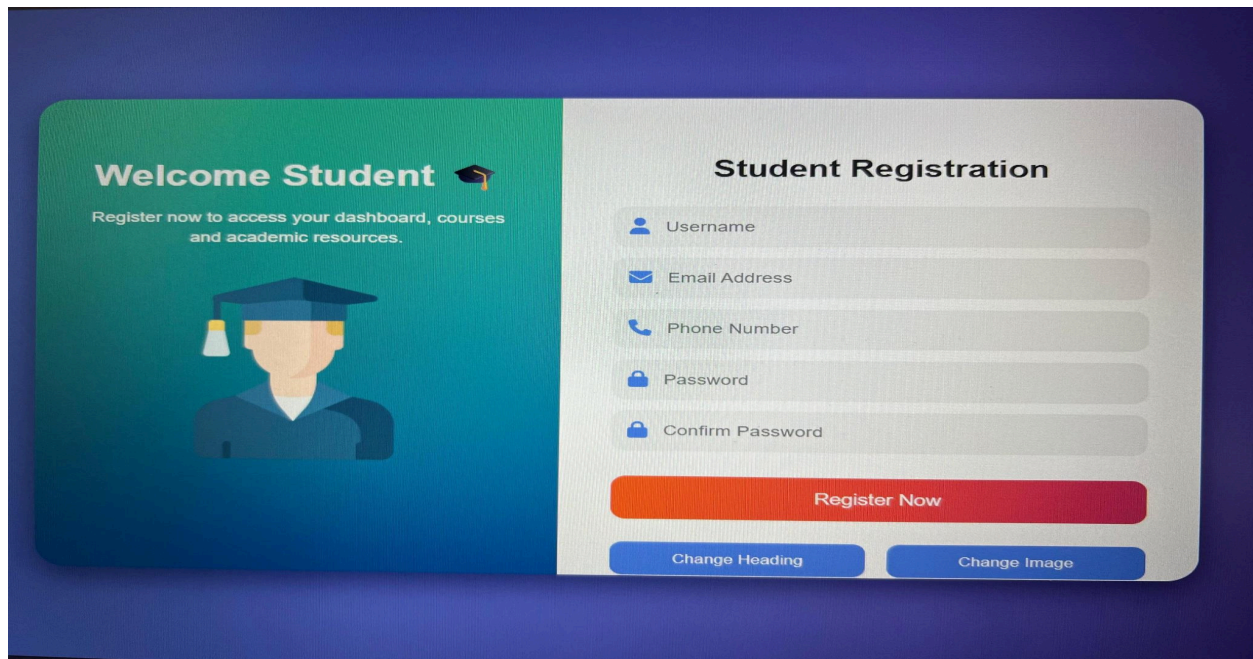

<script>
document.getElementById("myImage").setAttribute("src", "new.jpg");
</script>
```

3. What is jQuery Ajax?

jQuery AJAX refers to the use of AJAX (Asynchronous JavaScript and XML) techniques through the jQuery library to send and receive data from a server without reloading the entire web page. It helps create faster, more interactive, and dynamic web applications.

Using jQuery AJAX, developers can easily perform tasks such as loading data, submitting forms, and updating webpage content asynchronously with simple methods like `$.ajax()`, `$.get()`, and `$.post()`.

Output: Screenshots of the output to be attached.



```
Last login: Fri Feb 6 19:16:32 on ttys000
arhaan@Arhaans-MacBook-Pro ~ % cd ~/Documents/arhaan_FSDL

arhaan@Arhaans-MacBook-Pro arhaan_FSDL % git checkout main
git pull origin main

Already on 'main'
Your branch is up to date with 'origin/main'.
From https://github.com/arhaanmulani/arhaan_FSDL
* branch          main          -> FETCH_HEAD
Already up to date.
arhaan@Arhaans-MacBook-Pro arhaan_FSDL % git checkout -b assignment3

Switched to a new branch 'assignment3'
arhaan@Arhaans-MacBook-Pro arhaan_FSDL % git branch

    assignment2
* assignment3
  main
arhaan@Arhaans-MacBook-Pro arhaan_FSDL % mv "/Users/arhaan/MIT /LABORATORY/ARHAAN_FSDL3" .

arhaan@Arhaans-MacBook-Pro arhaan_FSDL % ls
```



```
assignment2
assignment3
main
Arhaan@Arhaans-MacBook-Pro arhaan_FSDL % mv "/Users/arhaan/MIT /LABORATORY/ARHAAN_FSDL3" .

Arhaan@Arhaans-MacBook-Pro arhaan_FSDL % ls
ARHAAN_FSDL3  file1.txt  file2.txt  file3.txt  LICENSE
Arhaan@Arhaans-MacBook-Pro arhaan_FSDL % git add ARHAAN_FSDL3

Arhaan@Arhaans-MacBook-Pro arhaan_FSDL % git commit -m "Assignment 3: Added code folder"

assignment3 4c79bda] Assignment 3: Added code folder
3 files changed, 156 insertions(+)
create mode 100644 ARHAAN_FSDL3/index.html
create mode 100644 ARHAAN_FSDL3/script.js
create mode 100644 ARHAAN_FSDL3/style.css
Arhaan@Arhaans-MacBook-Pro arhaan_FSDL % git push origin assignment3

Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 14 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 2.05 KiB | 2.05 MiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Remote: Resolving deltas: 100% (1/1), completed with 1 local object.
Remote:
Remote: Create a pull request for 'assignment3' on GitHub by visiting:
Remote:   https://github.com/arhaanmulani/arhaan_FSDL/pull/new/assignment3
Remote:
To https://github.com/arhaanmulani/arhaan_FSDL.git
* [new branch]      assignment3 -> assignment3
Arhaan@Arhaans-MacBook-Pro arhaan_FSDL %
```

Problem Statement:

Write a program to design Student registration form by using HTML, CSS having following fields: Username, Email, Phone number, Password, Confirm Password and write external javascript code to achieve following validations

- Fields should not be empty. If spaces are entered those should be considered empty
- Phone number must accept only numeric values and it should be 10 digits
- Password length must be at least 7 and it should contain at least one capital letter, one digit and one special character from the set (&,\$,#,@)
- Value entered in password field and confirm password fields must match

Email address must contain @ sign and a ., there should be few letters before the @ sign, there should be three letters between @ sign and a . There must be 3 or 2 letters after the . (hint: Use regular expression)

Write a client-side script with JavaScript to access and manipulate Document Object Model (DOM) objects in an HTML web page. Develop a dynamic web page using javascript and DOM. Make use of the following for accessing elements

- getElementById, getElementsByTagName, getElementsByClassName
- Change the text using innerHTML property
- Change the CSS properties like color, position of a particular element on the page
- Change the image source after clicking on a button
- Add a text node and attach it to a parent node
- Delete a node

Include jQuery to perform following operations:

- Change button text using jQuery.
- Set background-image using jQuery CSS property.
- Access HTML form data using jQuery.
- Add attribute using jQuery

Use this reference link for jQuery :

<https://www.w3resource.com/jquery-exercises/part1/index.php>